

LECTURE 21: MATH1061/7861
TOPICS: COMPOSITION OF FUNCTIONS
DATE: THURSDAY APRIL 16, 2026

Learning objectives:

(1) Reinforce the definitions of one to one and onto functions.

(2) Learn how to compose functions.

Feedback from class survey

* Videos are mandatory.

Reading from textbook is optional.

* Not necessary to go through pre-lecture notes prior to class.

* Concerns about level of difficulty: (Average Exercise Sheets ~ 87%.

↳ Mandated by the program

↳ Some challenging bits will be made not assessable.

→ I will give precise instructions for the final.

Q1: Define two functions f and g from natural numbers to itself by

$$\begin{aligned}f(n) &= n + 1, \\g(n) &= n^2.\end{aligned}$$

(a) Write down explicit formulas for

$$f \circ g \quad g \circ f \quad f \circ f \quad g \circ g.$$

$$(f \circ g)(n) = f(g(n)) = f(n^2) = n^2 + 1$$

$$(g \circ f)(n) = g(f(n)) = g(n+1) = (n+1)^2 = n^2 + 2n + 1$$

$$(f \circ f)(n) = f(f(n)) = f(n+1) = n+2.$$

$$(g \circ g)(n) = g(g(n)) = g(n^2) = (n^2)^2 = n^4.$$

(b) Let $f^{(k)} = \overbrace{f \circ f \circ \dots \circ f}^{k\text{-times}}$. Write explicit formulas for $f^{(k)}(n)$ and $g^{(k)}(n)$.

$$f^{(k)}(n) = f(f(\dots(f(n))\dots)) \\ = n + k.$$

$$g^{(3)}(n) = g(\underbrace{g(g(n))}_{n^4}) = (n^4)^2 = n^8$$

$$g^{(4)}(n) = g(g^{(3)}(n)) = g(n^8) = (n^8)^2 \\ = n^{16}$$

$$g^{(k)}(n) = \cancel{(n^2)^k} (n)^{(2^k)}$$

Q2: Let $f : \mathbb{Z} \rightarrow \{0, 1, 2, 3, 4\}$ be defined by

$$f(n) = n^2 \pmod{5}.$$

← means we take remainder modulo 5.

(a) Compute $f(7)$, $f(13)$ and $f(2026)$.

$$f(7) = 49 \pmod{5} = 4$$

$$f(13) = 169 \pmod{5} = 4$$

$$f(2026) = 4,104,676 \pmod{5} = 1$$

(b) What is the pre-image of 1?

$$f^{-1}(1) = \{n \in \mathbb{Z} \mid n^2 \equiv 1 \pmod{5}\}$$

$$= \{n \in \mathbb{Z} \mid n \equiv 1 \text{ or } 4 \pmod{5}\}$$

$$\begin{aligned} \cancel{f(2026)} \quad 2026^2 \pmod{5} &= 2026 \pmod{5} \times 2026 \pmod{5} \\ &= 1 \times 1 = 1 \end{aligned}$$

(c) Is f one-to-one?

$f(7) = f(13) \Rightarrow f$ is not
one to one.

(d) Is f onto? No. ~~2~~ 2 and 3 are not in
the image.

~~$f(0) = 0$~~
 ~~$f(1) = 1$~~

